PROJECT OVERVIEW

Power Glove Vision Project Overview

Architecture, controls, security, deployment, and project status.

2026 EDITION / IAIN BENNETT / MIT LICENSE

Current project version: 0.4.0; public candidate: v0.4.0-rc.1. This candidate promotes the tested MediaPipe efficiency work: newest-frame capture, four inference threads, 30-fps-first camera negotiation, off-thread lightweight preview rendering, and Latest-coordinate native movement with guarded reacquisition. Optional, capability-checked camera latency and exposure choices are available without changing the compatible defaults. Update the Controller and RetroPie together using the installation guide.

PowerGlove Vision lets you play RetroPie games by moving your hand in front of a camera connected to the **PowerGlove Vision Controller**, built on an Arduino UNO Q. Use your bare hand or a plain glove; there are no glove electronics to build. The PowerGlove Vision Controller tracks your movements and sends controller input to a Raspberry Pi, where RetroArch sees a virtual gamepad named **PowerGlove Vision**.

The project includes eleven profiles: nine reusable Programs A-I and dedicated controls for Bad Street Brawler and Super Glove Ball. RetroPie can select a profile automatically when you launch a registered game. Glove Academy teaches hand movements and gestures through sixteen guided lessons, with camera feedback, saved progress for each player, and a **Glove Master** award for completing them all. Learning mode lets you practise without sending input to the cabinet. Its optional Pixel Pal-guided personalization wizard adjusts recognition to a player's hand without retraining the model, changing game mappings, or exposing raw thresholds during normal use. The local Play page adds a camera-controlled Rock Paper Scissors match against Pixel Pal without requiring RetroPie.

The Dashboard's **Start controller** choice is retained across Controller application and system restarts as an armed preference. Armed does not mean that controls are always being sent: a registered RetroPie launch maintains a short renewable game session only while RetroArch is running. Controller delivery resumes after a Controller restart when that session is still live, then returns to neutral when the game ends, the session becomes stale, or an unregistered game is launched. **Stop controller** remains sticky until the player explicitly starts it again. Manual Dashboard profile selection remains available for testing outside the registered-game flow.

The cabinet supports two Super Glove Ball paths. `lr-fceumm` is the complete, standard-joystick fallback and remains the safe default. The separately named `lr-nestopia-powerglove` core supplies native absolute X/Y and Z coordinates plus open-hand, closed-hand/fist, and index-point states. Exact-ROM traces and deterministic headless tests confirm their packet bytes alongside detection, Start, four-direction activation/release, small continuous movement, and safe neutralization. Live cabinet play now confirms every implemented Super Glove Ball action: native Start, open-hand release/throw, closed-hand grab/catch, index-point Robo-Bullet fire, and fist-plus-forward Power Punch. Corrected Y orientation and full-field X/Y movement are also playable. Movement still has noticeable latency to refine. Native wrist rotation and the remaining unused packet button fields stay neutral until exact-ROM testing gives them a purpose; they are not missing from the game actions confirmed in the completed session.

Choose **Setup -> Matrix attract mode** to keep the idle animation On, Dim it, or turn it Off except for faint connection pixels. This does not change game displays, T, L, or gesture recognition. The setting saves without a tracker restart. Off mode has separate app, console-service, authenticated-console, and Networking pixels; the fourth reports a physical Wi-Fi or Ethernet link, including USB dock Ethernet. Setup distinguishes unavailable telemetry from disconnection. Updating the host sampler enables Ethernet detection without a new matrix firmware format.

Setup starts with four labelled status markers matching the Off-mode pixels: Controller app, console service, authenticated response, and Networking. Green means confirmed, red means disconnected or not confirmed, and grey means unknown. Tracking, controller output, and the saved console appear alongside them. Both pairing methods require the approval PIN displayed on the Controller matrix.

Setup places **Players** below Controller status, followed by **Matrix attract mode**, **Connection and startup**, guided pairing, Games, and the optional statistics preference. Select **Save settings** before pairing; the three steps use the saved console address: choose a method, confirm the Controller certificate and matrix PIN, then enter the RetroPie code or SSH credentials. Both methods remain available. After selecting **Pair with RetroPie**, a visible **Pairing in progress** panel leads to **Pairing complete** or an actionable error. Expiry and submitted failures require fresh confirmation. Controller Start/Stop and shutdown are on Dashboard. **Check console address** tests name resolution; use a running game to verify delivery.

When a submitted pairing attempt finishes, the matrix releases the approval PIN and resumes its normal display. When idle, the glove animation follows your On, Dim, or Off attract setting; active game and status displays still take priority.

The Changelog retains the completed Setup review and release evidence instead of maintaining a second history. Player calibration, complete backups, background hostname refresh, independent Networking indication, signed controller sessions, and web-module cleanup are implemented. Controller transport requires matching version-2 software on both computers; follow the coordinated upgrade instructions.

Manage players and backups in **Setup -> Players**. Select the active player on Dashboard or in Glove Academy; that selection applies to both practice and gameplay. Dashboard combines the game name and session status in one Game card.

On Dashboard, **Center hand** saves the resting reference for the selected player. If that player needs centering, guidance appears beside the controls before you can start controller output.

Choose a guide

User manuals

You want to...	Read...
Install both devices and play your first game	Installation Guide
Choose a camera, frame rate, or exposure setting	Camera Guide
Find a command or connection reminder	Quick Reference
Learn a game's gestures and try a short challenge	Game and gesture guide
Choose or experiment with Programs A-I	Programs A-I manual
Join Pixel Pal's suspiciously well-fingered scavenger hunt	Game and gesture guide and Programs A-I manual
Recognize matrix animations and letters	Matrix display guide

Technical documentation

You want to...	Read...
Get the complete project at a glance	Project overview PDF
Understand components and data flows	Architecture
Understand joystick versus native glove input	Native emulation explained
Review Super Glove Ball packet and gameplay evidence	Native compatibility record
Change settings or look up command flags	Configuration Reference
Review measured native and FCEUmm direction response	Direction-response benchmark
Review movement-filter evidence and experiments	Motion smoothing analysis
Measure native X/Y or isolate it from Super Glove Ball behavior	Native movement validation
Understand network and pairing boundaries	Security policy
Change the project or its documentation	Contributing guide
Check dependency provenance or release history	Third-party notices and Changelog

Pixel Pal's Extra-Digit Hunt

Pixel Pal has discovered that a few illustrated gloves left the art department with a generous interpretation of hand anatomy. Naturally, this is now a game.

Count every illustrated hand showing five fingers plus a thumb. Count each appearance, even when the same artwork returns. Pixel Pal has tucked the answers at the back of the two illustrated guides and behind a reveal in built-in Help, so there are no spoilers here.

An automated documentation check keeps the answers synchronized with the art. The art itself remains untouched in the interests of arcade archaeology - and because it is far too funny to fix.

The web footer shows exact software and running matrix firmware identities. Glove Academy supports twelve player presets, saved lesson progress, and portable version-2 hand-setup backups containing name, personal and effective sensitivity, software identity, and per-player calibration. Selecting a player immediately loads their sensitivity, progress, and saved center, with output paused. Use **Center hand** for new players or after changing the physical setup. Version-1 portable backups are no longer accepted. Navigation and controls adapt to phone and tablet widths.

Quick start

Prepare the PowerGlove Vision Controller with Arduino App Lab and use an existing RetroPie installation. Connect both to the same trusted network and attach the camera through a powered USB hub. Keep a physical controller available for RetroArch setup.

The commands in the Installation Guide select the latest published stable release. Use the same release on both devices. Download `install-uno-q.sh` and `install-retropie.sh` from that published release. Run the first on the Controller and the second on RetroPie as your normal login user. Each verifies its package and requests sudo access when needed. The UNO installer includes the Arduino sketch, early-start helper, shutdown helper, and guarded USB-camera recovery helper; no separate App Lab import or helper installation is needed.

Follow the Installation Guide for copyable commands, pairing, calibration, and your first game. Both scripts also support `--check` and repeatable updates while preserving personal settings. The release-owned `config/profiles.json` baseline is backed up and replaced, while calibration and personal tuning under `data/` remain in place. Installer assets must be published before the release download commands become available.

When gestures are off, the matrix plays a lightning-and-glove animation with curling fingers, a travelling spark, and a soft grayscale glow. See the Matrix display guide; the revised loop requires updated matrix firmware.

The tested shared baseline includes responsive `0.28` activation and `0.14` release thresholds, calibrated native X/Y reach with an 8% edge margin, and **Latest coordinate** as the native movement default. It sends each newest valid, reach-clamped palm position directly during continuous tracking. After a brief MediaPipe dropout, one contradictory or unusually distant reacquisition may be held for the next fresh result; strongly aligned forward movement remains immediate. This guard does not predict, smooth, or overshoot. Latest coordinate is the only live native X/Y behavior. The selected camera frame's capture time and MediaPipe Hands remain authoritative for palm landmarks, finger curls, and gestures. The earlier bounded and optical-flow experiments are archived in the source tree for research and are no longer live movement options. These are suitable starting values for every installation. A neutral calibration is different: it records the palm center, apparent hand size, wrist angle, and resting jitter for one camera and playing position, so the installer never substitutes another person's recorded coordinates for yours.

Install the same release on the Controller and RetroPie. Automatic game-session resume depends on the current Controller worker and current RetroPie launch hook being present together; mixed old/new installations continue to fail safe but cannot provide the renewable session behavior.

When a registered Super Glove Ball ROM is present, the RetroPie installer offers to build the optional native core locally from pinned GPLv2 Nestopia source. If you decline, nothing changes and FCEUmm remains available. If you accept, both cores appear in RetroPie's per-ROM launch menu; FCEUmm stays selected until you choose the native entry. The project does not distribute ROMs or a compiled Nestopia core in its ordinary installation archive.

Controls

Calibration records the resting hand position that the app treats as the centre of movement. Move away from that position to give a direction and return to it to release that direction. Recalibrate after moving the camera or changing your playing position. Direction activation and release are shared by all FCEUmm profiles and automatically rise above measured resting-hand jitter; you do not normally calibrate each direction. Some profiles replace ordinary hand movement with wrist steering or other controls, as shown below.

Calibration accepts 24 geometrically valid hand observations. MediaPipe Hands' reported score identifies handedness certainty, not landmark or position confidence, so it is shown diagnostically but is not used as a false quality gate. The whole hand must still be detected with usable palm geometry. Holding the same neutral pose at the same distance should reproduce a very similar reference, but ordinary tracking variation means the saved numbers will not be identical. A completed calibration is saved atomically and reused across games and restarts; an incomplete attempt does not replace the previous file.

Across the profiles, hold a **V sign** steadily for half a second to send Start and a **thumbs-up with the other fingers closed** to send Select. These poses suppress A/B attacks; some profiles can still generate directional or auxiliary input, so keep your hand near its resting position while using them. Start sends only one press and must see a clearly non-V pose for 0.30 seconds before it can trigger again.

Programs A-I

These reusable mappings produce ordinary NES controls. You do not need to launch Bad Street Brawler first. The table describes controller output; its effect depends on the game. A pulsed button repeatedly presses and releases.

Program	Movement	Actions and special gestures
A - Pinball	Ordinary movement is disabled.	Index curl sends A; thumb curl sends Up; wrist roll sends B. Pulling back toggles combined flippers.
B - Joust	Move your hand left or right.	Index or middle curl pulses A; thumb curl holds B.
C - Gyruss	Roll your wrist left or right.	A straight index finger holds A; pulling back sends B. Use the game's Attack Control B mode.
D - Challenge	All four hand-movement directions are reversed.	Thumb curl sends A; index curl sends B.
E - Defender II	Move your hand in four directions.	Thumb curl sends A; wrist roll sends B; ring-finger curl rapidly alternates left and right.
F - Sesame Street	Ordinary directional output is disabled.	Moving an open hand away from centre sends A; closing all fingers sends B.
G - Gun Smoke	Move your hand in four directions; wrist roll adds left or right.	Index curl sends A; pushing forward sends B. Combine them for A+B. Menu guard suppresses all ordinary controller output.
H - General	Move your hand in four directions.	Index curl pulses A; thumb curl pulses B.
I - Knight Rider	Roll your wrist to steer; lower your hand to brake.	Index curl sends Up for acceleration; pushing sends Up+A for turbo; thumb curl sends B.

Programs A, D, and H have no default ROM assignment. Choose one on Dashboard to try it, then register the exact game filename if you want automatic selection. The [Programs manual] includes illustrations; the [Game and gesture guide] adds objectives and tips.

Bad Street Brawler

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your thumb	Pulsed B
Curl your middle finger	A+B
Roll your wrist left or right	A plus that direction
Push toward the camera	Glove Zap: short simultaneous Left + Right pulse

Push toward the camera for Glove Zap, then return to your starting distance before trying again. A push or pull must cross its threshold on two consecutive fresh observations and travel at least 0.10 palm-scale units in the intended direction within 250 ms. This rejects a one-frame scale jump and a stationary hand that merely begins near or far from the camera. Bad Street Brawler needs its game-specific emulator setting; see the configuration reference.

Super Glove Ball

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your index finger	A
Curl your thumb	B

The same responsive mapping remains the explicit FCEUmm fallback. Headless tests using the same exact ROM show that both paths visibly activate and release every direction by frame 3. Their semantics differ: FCEUmm supplies held digital directions, while the native core supplies an absolute target position. The native path has passed exact-ROM detection, Start, continuous X/Y, absolute Z, open/fist/index packet, and safe-neutralization tests. Live full-game play confirms grab/throw, index fire, and fist-plus-forward Power Punch. Native movement uses per-player reach calibration and the selected MediaPipe response mode. A brief missed observation holds only the last X/Y coordinate for up to 120 ms, while actions release immediately. On recovery, Latest accepts the new measurement immediately unless it contradicts established motion or is an unusually distant non-forward jump; only that questionable result waits for one fresh confirmation. A longer loss neutralizes the native sample. Wrist rotation and remaining unused native packet fields stay neutral. See the native compatibility record.

The eight-ROM input audit confirms that the other listed games consume standard NES controller bits. They continue to use FCEUmm and the same global recognition settings.

For movement-latency investigation, the baseline procedure collects fresh timing observations without changing camera settings or controls. It keeps Controller software timing separate from network, emulator, and display delay. The optional PowerGlove Vision Controller dot test reuses the cabinet's installed `lr-powerglove-dot` core to display the same receiver X/Y publication without game movement logic, with read-only input-range and validity measurements.

The native latency session tools guide stationary/movement windows, optionally correlate software traces, and extract annotated evidence from an original hand-and-screen recording. They are disabled during normal play. A privacy-safe preflight records the exact two-device test state without changing it; smoke mode checks framing before the full session, and the reversible trace helper restores production services before exporting its bounded evidence. Historical trace tools can compare recognized, optical-flow, selected, and filtered coordinates without recording video; the current live path reports the newest valid MediaPipe coordinate directly. Physical hand-to-screen latency still requires synchronized recording.

The production Controller runs the proven CPU MediaPipe Hands path at 640×480, with four explicitly selected inference threads, a `0.35` tracking-confidence threshold, and a `2.25` next-frame hand search area. The larger search area recovered five of nine previously missed fast-sweep frames in repeatable replay without a material latency or false-activation cost. Camera rate defaults to Automatic, which tries the measured 30 fps path before safely accepting the driver's supported rate. An isolated Adreno GPU probe successfully created a delegate but the first MediaPipe Tasks graph was much slower than production, so no GPU wheel or runtime change ships in this candidate. A lean, output-paused GPU palm/landmark experiment remains research.

Use the web interface

Setup includes **Joystick dead zone**, saved separately for each player. Small requires less hand movement to press a direction; Large gives more room around center. **Use standard size** selects the existing 0.28 activation / 0.14 release pair; select **Save dead zone** to apply. The slider sets all four directions together, without changing center or native Super Glove Ball reach. Live direction indicators work while tracking is active. Separate directional thresholds remain under Glove Academy -> Tune gestures -> Advanced thresholds and diagnostics.

The Controller website uses the logo's hand-and-target emblem for browser tabs and saved home-screen shortcuts.

Page	What it does
Dashboard, `/dashboard`	Shows the camera and generated inputs; selects the current profile and starts or stops delivery.
Play, `/play`	Runs a camera-controlled Rock Paper Scissors match against Pixel Pal, with cabinet input paused.
Glove Academy, `/learn`	Provides sixteen mapping-independent practice lessons and guided gesture tuning, with game input paused. Player presets retain individual sensitivity, progress, and the Glove Master award across restarts. Select the same active player used for gameplay; manage players and hand-setting backups in Setup.
Help, `/help`	Opens the local manuals and PDFs; **This console** shows current connection details.
Setup, `/setup`	Saves connection, camera, and startup settings; its camera dropdown lists Automatic and discovered usable cameras. The Games section edits RetroPie mappings with backup and restore. Pairing requires HTTPS on port 8443.

With **Gestures off** selected, the camera stays closed. Choose an active profile, open Play, or open Glove Academy to begin. Wait for the camera view before practicing or playing; starting immediately after a reboot can take longer.

The live camera preview is diagnostic rather than part of controller output. **Show statistics** is off by default and can be enabled on Dashboard or Setup; the browser remembers the choice. When it is off, the Dashboard does not render or retain the optional controller, axes, finger, performance, or event panels, and the worker skips their derived housekeeping. UI-visible control transitions are published immediately while routine detail refreshes at about 10 Hz through a latest-only status worker. Tuning locks are resolved before inference and no Dashboard lock is taken between completed inference and UDP transmission. Camera capture continuously keeps only the newest frame, and browser JPEG encoding runs on a separate latest-preview worker that may drop stale preview jobs. Preview mirroring, annotation, and encoding use a 320×240 copy during gameplay, while Academy and tuning keep the full preview. The preview remains capped at 5 fps. Closing Dashboard or Glove Academy while playing a RetroPie game still avoids optional drawing and encoding work; tracking and controller delivery continue.

For repeatable engineering checks, `scripts/benchmark-post-inference.py` runs a camera-free signed-send and Dashboard-housekeeping soak. The saved-clip replay still measures MediaPipe recognition separately; neither headless tool claims physical hand-to-display latency.

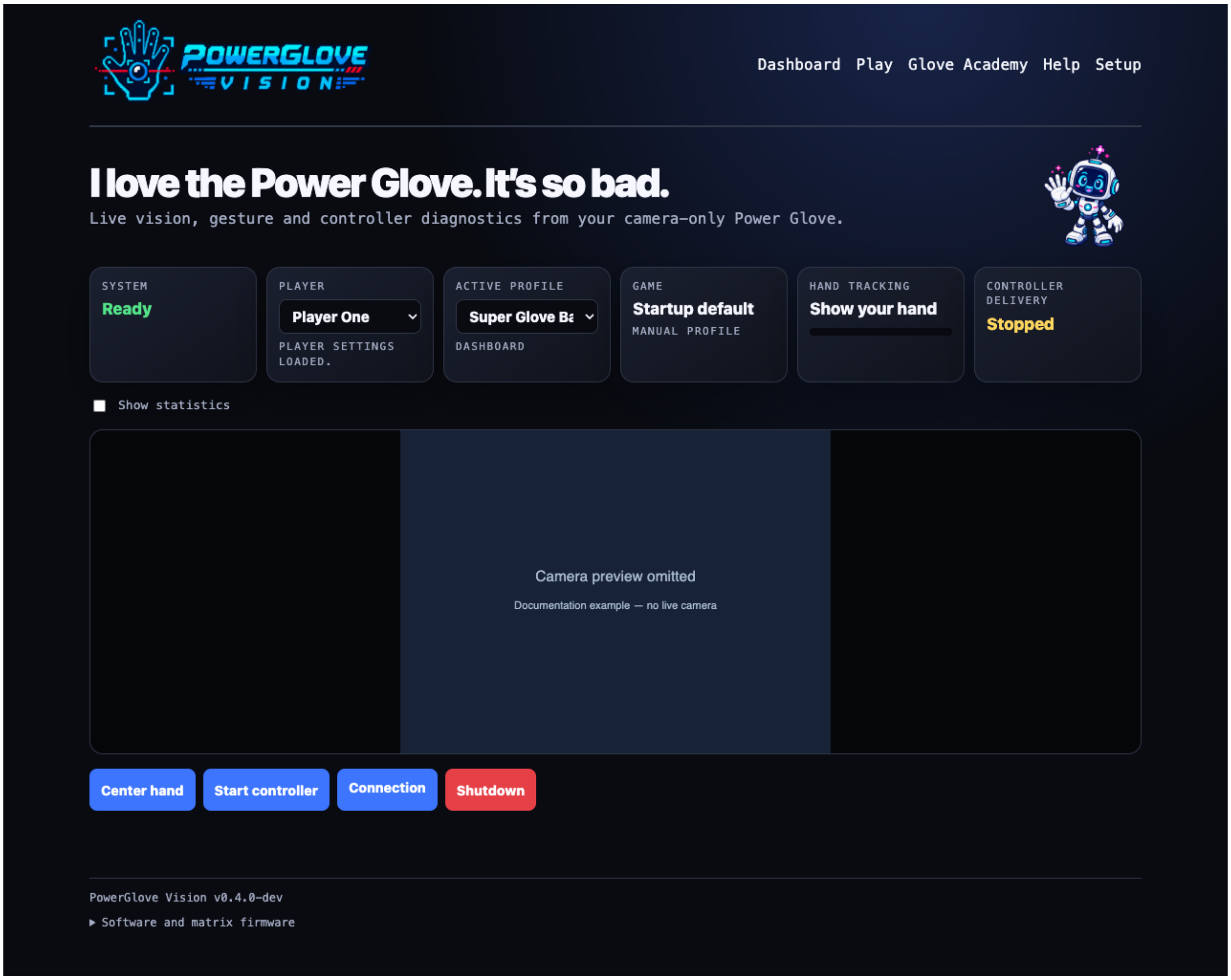
Strong light behind the player can leave the hand dark even when the room looks bright. Prefer light from the camera side or move bright windows out of the background. The project does not force hardware backlight compensation: on the tested Razer Kiyo Pro it made the measured backlit scene darker.

Setup also offers opt-in camera controls. **Low latency - Direct V4L2** reads the newest Linux MJPEG driver buffer and automatically falls back to OpenCV if the camera or negotiated format is incompatible. **Razer Kiyo Pro - tested low latency** keeps automatic exposure, requests a fixed frame rate using advertised standard UVC controls, and also requests the Kiyo's volatile HDR-off mode. **Manual exposure and gain** is available only with Direct V4L2. The Controller first checks the camera's advertised controls and reports the values actually applied. Unsupported settings fall back visibly to automatic exposure. Automatic remains the portable installation default; this project's Kiyo Pro tested slightly more reliably at exposure `78` and gain `96`, without a measured latency difference. Manual controls are restored to automatic when the camera is closed, while the saved preference is reused the next time vision starts.

The direct reader passed a live compatibility check on this project's Kiyo Pro and exposed valid driver sequence numbers and monotonic timestamps. It remains an option rather than a universal default because other camera drivers may not provide the same Linux MJPEG interface. A matched lean MediaPipe output test did not produce a meaningful end-to-end improvement, so the complete proven graph remains selected.

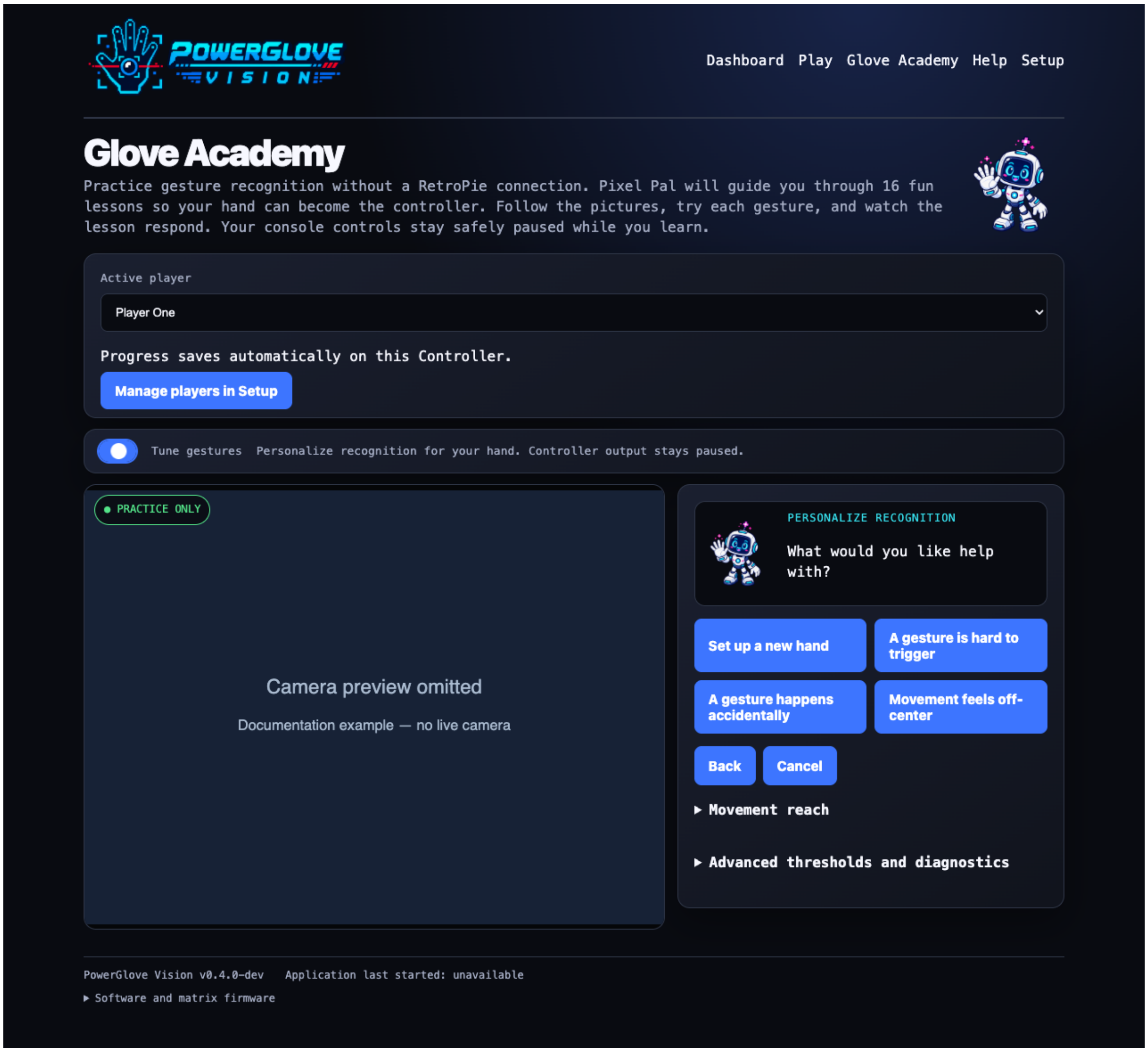
The Controller host helper supports one UVC camera. Installation works with or without the camera connected. On the first healthy sighting it records the camera and its actual parent USB hub in a root-owned allowlist, disables autosuspend for both, and automatically updates that association if the camera is later moved to a different hub. If the camera remains missing for 15 seconds while vision is requested, PowerGlove Vision makes one guarded recovery attempt for that outage by resetting only the last successfully observed hub. This can briefly interrupt USB Ethernet; Wi-Fi remains available. If a camera has never been seen-or does not return after that attempt-reconnect or power-cycle it and check the hub and cable rather than repeatedly resetting it.

Stop controller pauses delivery while leaving active tracking available. **Gestures off** closes the camera. **Shutdown** requests a Linux halt, but the tested Arduino UNO Q hardware restarts afterward. A disappearing website is not proof that it is safe to remove power. See the installation guide before using Shutdown.



Dashboard showing the selected profile and controller readings

The screenshots below show the current interface with isolated sample data. Camera imagery is replaced with a labelled placeholder for privacy.



Glove Academy with Pixel Pal guiding the personalization choices

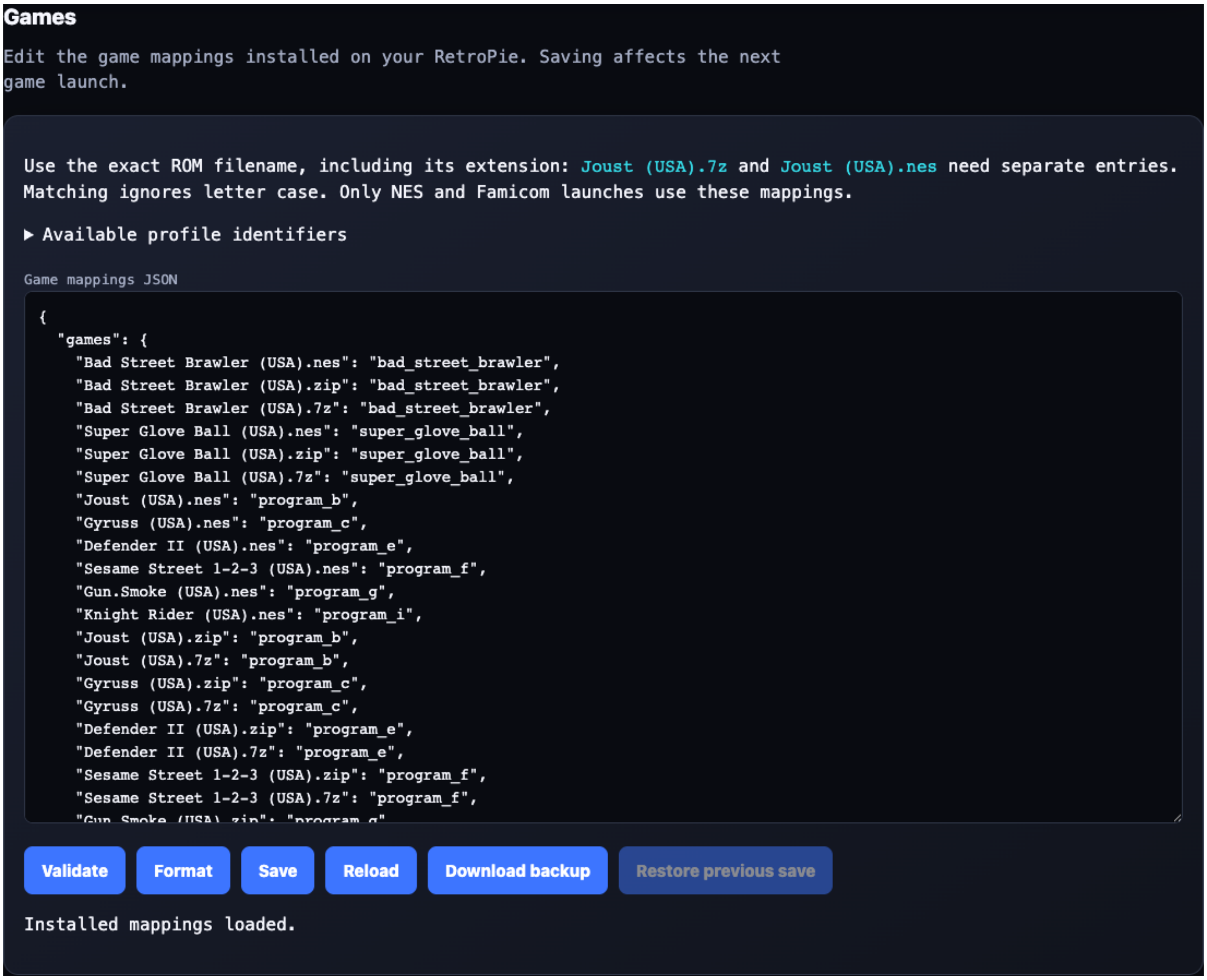
Learn and practise: open **Glove Academy** and choose your player to work through sixteen lessons, from showing and centering your hand to movement and gesture control. Follow the illustrated instructions and camera feedback, practise one movement at a time, and return later to your saved progress. Complete every lesson to earn **Glove Master**. Lessons teach the gestures independently of the selected game mapping; use the Gameplay Guide to see what those gestures do in each game. The Controller displays a scanning **L** during learning mode, with cabinet input paused.

Choose each player in turn and select **Back up hand setup** to download a separate file named for that player, such as `iain-powerglove-hand-setup.json`. Your browser saves it on the computer, phone, or tablet you are using, usually in **Downloads** or the folder you choose. To restore, select the player you want to update, choose **Restore hand setup**, and pick that player's saved file from your device. Review it before confirming; restore updates the selected player, rather than adding a new one.

Personalize recognition: switch on **Tune gestures** when recognition needs adjustment for your hand. This optional mode displays a scanning **T** and also pauses cabinet input.

Pixel Pal first asks what feels wrong, then presents one instruction at a time. Recording starts only after the whole hand has been tracked clearly and steadily; the user presses **I'm ready** and sees a countdown. The wizard previews a conservative adjustment, requires two successful uses and releases plus three neutral seconds, and enables Save only after that check passes. Saved recognition settings apply across profiles. Numerical thresholds, selective reset, manual preview, and the private diagnostic capture live under **Advanced**. Diagnostic video remains on the Controller, is deleted after analysis or cancellation, and its downloadable aggregate report contains no pictures or per-frame hand data.

The separate **Movement reach** section exposes the selected player's left, right, up, and down spans. These are normalized distances from the saved center; smaller values require less physical travel. Its summary shows the resulting tracking-area dimensions and aspect ratio. **Save reach values** changes only those four spans, while **Restore full camera field** returns all four to the camera-boundary default without changing center or gesture thresholds.



Games editor in the lower part of Setup

Scroll down **Setup** to **Games** to map exact ROM filenames to profiles. Saving affects the next game launch, not the game already running.

New builders can start with [Build your own: parts, cost, and difficulty](). For the game-input background, read [How native Power Glove emulation works](). When something fails, use [Troubleshooting by symptom](). All three are available in Controller Help and as printable PDFs. Help orders user manuals from console details through game controls, programs, matrix displays, building, installation, and troubleshooting. Technical documentation starts with the project overview and architecture before configuration and detailed evidence.

Maintain or extend the project

Use the Installation Guide's maintenance section for updates, and the Configuration Reference for settings and all command options. The Contributing guide covers tests, documentation, package verification, and releases. Printable editions are stored in output/pdf/. The maintained Markdown set is intentionally consolidated into 19 documents: operational details live with their owning guide, and third-party licensing, provenance, asset origins, and native-core modifications share one notice. Regenerate PDFs only after the Markdown review is complete.